



Tom Petersen,
Intel Fellow

Talking XeSS

All the juicy details about XeSS that you wanted to know

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Intel is already making waves with their own Super Sampling tech, Intel XeSS. With their Arc GPUs right around the corner, we sat down to have a chat with Tom “TAP” Petersen, Intel Fellow to know what really makes XeSS tick.

Q How open is Intel XeSS? Is it similar to what AMD has with their GPUOpen Platform?

It's hard (for me) to say exactly what AMD is doing with a hundred per cent certainty, but I can say what we're doing. We view XeSS as an open technology. We've made our SDK basically open and we're making a reference version of the XeSS implementation in HLSL (High-Level Shader Language), which is the shader language. So somebody could take that and optimise it for their platform. We're also doing an Intel-only version of XeSS with custom

fused cores and custom fused layers to get the optimum performance. That's where we're focusing most of our energy right now. Eventually, we'll have a DP4a version which will run on integrated graphics and as that happens, we'll start to push that out to become more of an open standard. I think that there's always going to be a question about, you know, what's the home for that going to be? Just like most of what is open, it depends how good it is and how effective it is. Also on whether it gets used everywhere, etc. and that's still to be determined.

Q Could you shed some light on how much of an impact the XMV and DP4a instruction sets have? Because that's what sets the Arc GPUs apart from anything else that can run XeSS, right?

Yeah, and that's the key part. We're doing fused kernels for XMV so think of it as the basic network being a CNN which is a well-understood architecture and we're using a popular topology that's already been validated. We're retraining that and then perform infused kernel optimisation which is like assembly code for graphics cards. We're doing all that to get the execution time for the XMV path down to as low as possible. That's like a finely crafted work of art. We're not doing that right now on our HLSL version but we will be doing that for our DP4a version. So, we are going to focus most on the XMV version and there's work to be done yet to make the DP4a version great.

Q When you're training the neural net to get better with time? Are you training it on a per-title basis?

I know that you're not releasing updates on a per-title basis and there aren't per-title optimisations coming out but you'll still be improving the tech, right?

We'll definitely be improving the tech and we'll be releasing the tech as versions but not really per-title. Games that take a particular snapshot of it will usually not automatically take the next one. They'll probably stay where they are. So we're